

Generation Interconnection System Impact Study Report

For Queue Project Q18-0945

Blue 138 kV

July 25, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0945. The Interconnection Customer proposes a 452.5 MW natural gas combined cycle facility interconnecting at the Blue 138 kV substation in the Virginia area. The request was received on February 20, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 353 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$58,765,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rochester - Radford Army Ammunition Plant 138.0 kV L	Lowoor - Hinton Ap 345.0 kV	160.5	183	5.4
Coyote - Hobart 138.0 kV	Clinch Riv - Carlton Hills 345.0 kV	153.8	253	20.0
Rochester - Sysquoc 138.0 kV	Hinton Ap - Lowmoor 345.0 kV L	148.4	209	5.4
Ingles St - Svp 138.0 kV line	Hinton Ap - Lowmoor 345.0 kV	145.1	191	5.4
Svp - N Claytor 138.0 kV	Lowmoor - Hinton Ap 345.0 kV	143.6	193	5.4
Taloga - Pomona Heights Substation 138.0 kV line	Hinton Ap - Lowmoor 345.0 kV	142.7	252	5.4
North Claytor - Radford Army Ammunition Plant 138.0 kV L	Lowmoor - Hinton Ap 345.0 kV	140.6	211	5.4
Lake Forrest - Sysqoc 138.0 kV	Lomoor Substation - Hinton Ap 345.0 kV Line	133.5	243	5.4
Lytle - Lake Forrest 138.0 kV	Lowmoor - Hinton Ap Substation 345.0 kV line	131.2	256	5.4
Winnebago Wind Power Project - Lebanon 138.0 kV	Barr Lake - Kean Mtn And Grassy Creek 345.0 kV	130.0	242	6.6
Lytle - Pomona Hts 138.0 kV Line	Hinton Ap - Lowmor 345.0 kV	129.0	270	5.4
Byhalia - Lebanon 138.0 kV Line	Kean Mtn And Grassy Crk - Barr Lake 345.0 kV	128.7	235	6.6
Haysi Substation - Leesburg Picciola Substation 138.0 kV	Atoka - Carlton Hills 345.0 kV	127.8	269	7.2
Byhalia - Calpeak Pe 7 138.0 kV L	Kean Mtn And Grassy Crk - Barr Lake 345.0 kV	127.1	231	6.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Orovada - Carlton Hills 138.0 kV Line	Atoka - Carlton Hills 345.0 kV L	127.0	282	9.5
S. Land Park & 35Th Ave - Pringle 138.0 kV	Manatee - New Hope Ap 345.0 kV	126.1	216	6.6
Winnebago Wind Power Project Substation - Clinch Riv 138.0 kV	Barr Lake - Kean Mtn And Grassy Crk 345.0 kV	119.2	273	6.6
Claytor - Ingles St 138.0 kV	Hinton Ap - Lowmoor 345.0 kV	118.7	216	5.4
Red Hills - Beardsley St 138.0 kV	Lowmoor - Hinton Ap 345.0 kV L	118.6	507	7.9
Red Hills - Vaseaux Lake 138.0 kV line	Lomoor - Hinton Ap 345.0 kV	117.8	519	7.9
Mcclain Energy Facility - Vaseaux Lake Substation 138.0 kV Line	Hinton Ap - Lowmoor 345.0 kV	117.7	522	7.9
Halls Ridge - Pringle 138.0 kV	New Hope Ap - Manatee 345.0 kV line	112.8	243	6.6
S Princeton - Halls Ridge 138.0 kV	Manatee - New Hope Ap 345.0 kV	110.3	269	6.6
Daffin Gin - Calpeak Pe 7 Substation 138.0 kV Line	Kean Mtn And Grassy Creek - Barr Lake Substation 345.0 kV	108.8	264	6.6
Carlton Hills 138-1 kV Transformer Bank	Atoka - Carlton Hills 345.0 kV	108.6	401	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Radford Army Ammunition Plant - Rochester 138.0 kV	Rebuild line with higher-capacity conductor	\$9,730,000	29% (\$2,819,000)
Coyote - Hobart 138.0 kV	Rebuild line with higher-capacity conductor	\$6,686,000	57% (\$3,816,000)
Sysquoc - Rochester 138.0 kV	Rebuild line with higher-capacity conductor	\$5,435,000	29% (\$1,575,000)
Ingles St - Svp 138.0 kV	Rebuild line with higher-capacity conductor	\$1,661,000	29% (\$481,000)
Svp - N Claytor 138.0 kV	Rebuild line with higher-capacity conductor	\$3,932,000	29% (\$1,139,000)
Taloga - Pomona Heights Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$19,987,000	29% (\$5,790,000)
N Claytor - Radford Army Ammunition Plant 138.0 kV	Rebuild line with higher-capacity conductor	\$1,699,000	29% (\$492,000)
Lake Forrest - Sysqoc 138.0 kV	Rebuild line with higher-capacity conductor	\$4,522,000	29% (\$1,310,000)
Lytle - Lake Forrest 138.0 kV	Rebuild line with higher-capacity conductor	\$4,910,000	29% (\$1,422,000)
Winnebago Wind Power Project - Lebanon 138.0 kV	Rebuild line with higher-capacity conductor	\$2,842,000	39% (\$1,115,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lytle Substation - Pomona Hts 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,507,000	29% (\$436,000)
Lebanon - Byhalia 138.0 kV line	Rebuild line with higher-capacity conductor	\$9,005,000	39% (\$3,533,000)
Haysi Substation - Leesburg Picciola Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$25,134,000	24% (\$6,025,000)
Byhalia - Calpeak Pe 7 138.0 kV L	Rebuild line with higher-capacity conductor	\$35,161,000	39% (\$13,793,000)
Orovada - Carlton Hills 138.0 kV Line	Rebuild line with higher-capacity conductor	\$443,000	23% (\$101,000)
S. Land Park & 35Th Ave Substation - Pringle 138.0 kV	Rebuild line with higher-capacity conductor	\$5,888,000	25% (\$1,484,000)
Clinch Riv - Winnebago Wind Power Project 138.0 kV	Rebuild line with higher-capacity conductor	\$6,432,000	39% (\$2,523,000)
Claytor - Ingles St 138.0 kV	Rebuild line with higher-capacity conductor	\$3,588,000	29% (\$1,039,000)
Beardsley St - Red Hills 138.0 kV line	Rebuild line with higher-capacity conductor	\$12,343,000	13% (\$1,622,000)
Red Hills - Vaseaux Lake 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,674,000	13% (\$1,403,000)
Mcclain Energy Facility - Vaseaux Lake 138.0 kV	Rebuild line with higher-capacity conductor	\$9,268,000	13% (\$1,218,000)
Halls Ridge - Pringle 138.0 kV	Rebuild line with higher-capacity conductor	\$13,047,000	25% (\$3,288,000)
S Princeton - Halls Ridge 138.0 kV	Rebuild line with higher-capacity conductor	\$3,411,000	25% (\$860,000)
Daffin Gin - Calpeak Pe 7 138.0 kV	Reconductor line	\$930,000	39% (\$365,000)
Carlton Hills 138-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,904,000	23% (\$1,116,000)
	Total allocated to project		\$58,765,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.